

Introduction to probability

Probability spaces · Live writing edition

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Errors in a test sample

Suppose a fixed classifier makes 8 errors on 100 tested items. What has been measured?

A probability question about the same sample

A test sample has 100 items, 8 in error. If one item is selected uniformly, what is the probability it is an error?

A statistical question about future items

Do the 8 errors in the specified sample of 100 establish a future-item error probability of 0.08? State the inference assumption needed.

An inspection experiment

Inspect two components in order. Record P for pass and F for fail. List the possible records.

An event as a set of outcomes

For the ordered inspection outcomes PP, PF, FP, FF, list the event A: at least one component fails.

Exactly one failure

For outcomes PP, PF, FP, FF, let A be at least one failure. Find the event B of exactly one failure and compare it with A.

Intersection: both conditions

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cap D$.

Union: at least one condition

For ordered outcomes PP, PF, FP, FF, let C be first-component failure and D be second-component failure. Find $C \cup D$.

Complement: the remaining outcomes

For $\Omega = \{PP, PF, FP, FF\}$, let $A = \{PF, FP, FF\}$ be at least one failure. Find A^c .

Disjoint events

For the ordered inspection outcomes, let $B = \{PF, FP\}$ be exactly one failure and $E = \{FF\}$ be two failures. Determine whether they are disjoint.

Practice: two alarm indicators

Record 1 when an alarm is active, 0 otherwise, and use order for two indicators. List the outcomes in “at least one active,” “exactly one active,” and “neither.”

Assigning probabilities to outcomes

Suppose the ordered inspection outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Check that these are valid outcome probabilities.

Adding the probabilities in an event

The ordered outcomes PP, PF, FP, FF have probabilities 0.72, 0.08, 0.18, 0.02. Calculate the probability of exactly one failure.

Using the complement

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Calculate the probability of at least one failure directly and by the complement.

Deriving the complement rule

Explain why $P(A^c) = 1 - P(A)$.

Overlapping events in the inspection model

In the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$ and $P(FF)=0.02$. Let C be first-component failure and D second-component failure. Find $P(C)$, $P(D)$ and $P(C \cap D)$.

Deriving the addition rule

In the component-inspection model, C is first-component failure and D second-component failure. Use $P(C)=0.20$, $P(D)=0.10$ and $P(C \cap D)=0.02$ to find $P(C \cup D)$.

Practice: overlapping system alerts

Assume $P(A) = 0.20$, $P(B) = 0.15$ and $P(A \cap B) = 0.08$ for two system alerts. Find the probability of any alert.

Practice: exactly one system alert

Assume $P(A)=0.20$, $P(B)=0.15$ and $P(A \cap B)=0.08$ for two system alerts. Calculate the probability of exactly one alert.

Equally likely outcomes

A batch contains 20 items, of which 3 are defective. Select one item uniformly. Find the defect probability.

A die with unequal probabilities

Suppose a die has $P(\{k\}) = k/21$ for $k = 1, \dots, 6$. Find the probability of an even face.

Practice: checking a proposed model

Three mutually exclusive outcomes exhaust an experiment. Proposed probabilities are 0.2, 0.3 and 0.6. Is the model valid?

The sample space in a complete model

For one inspection, record 0 for pass and 1 for failure.
Specify the possible outcomes.

The collection of events

For the inspection recorded as 0 = pass and 1 = failure, list every subset of $\Omega = \{0, 1\}$.

Open the interactive experiment

The probability assignment

Suppose a pass/fail inspection has failure probability 0.10. Using 0 = pass and 1 = failure, assign a probability to every event.

The probability triplet

The example has outcomes $0 = \text{pass}$ and $1 = \text{failure}$, all four events, and failure probability 0.10 . Write the probability-space triplet.

Why the event collection must be closed

If we can ask whether A occurs, what related questions should the model also describe?

The probability rules

Complements let us ask what cannot occur; countable unions let us ask whether at least one of a list of events occurs. Once these questions are events, what rules should their probabilities obey?

Running frequency in a model

In the independent fair-coin model, the lab plots the running fraction of heads beside the model probability p . What can a finite run illustrate? What does it establish about a limit?

Open the interactive experiment

Practice: a batch inspection triplet

Ten labelled items contain two defective items, numbered 3 and 8. Select one item uniformly. Specify the triplet and the defect event.

Recording a numerical result

For the inspection model, the probabilities of PP, PF, FP and FF are 0.72, 0.08, 0.18 and 0.02. Let X count failed components. Assign X to each outcome.

The event $X = 1$

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find $P(X=1)$.

The distribution of the failure count

For the ordered inspection, X counts failures and $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$. Find its probability distribution.

A numerical set and its inverse image

For PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. For $B=\{1,2\}$, find $X^{-1}(B)$ and $P(X \in B)$.

An inverse image need not be an inverse function

On $\Omega = \{PP, PF, FP, FF\}$, X counts failures. Find $X^{-1}(\{1\})$ and explain whether an outcome can be recovered uniquely from $X=1$.

A random variable and a probability measure

For outcomes PP, PF, FP, FF with probabilities 0.72, 0.08, 0.18, 0.02, let X count failures. Find $X(\text{PF})$ and $P(\{\text{PF}, \text{FP}\})$. What does each map take as input?

Practice: a repair-cost random variable

For the model $P(PP)=0.72$, $P(PF)=0.08$, $P(FP)=0.18$, $P(FF)=0.02$, let X count failures and $Y=500X$ rupees. Find $P(Y>500)$.

A uniform measurement model

Let E be a uniform measurement error on $[-2, 2]$ mm. State how to calculate the probability of an interval $[a, b]$ contained in this range.

Meeting a tolerance

Assume measurement error E is uniform on $[-2, 2]$ mm.
Find the probability that the absolute error is at most 0.5 mm.

A uniform source variable

Let U be uniform on $(0,1)$. Find $P(U < 0.25)$ and $P(U > 2/3)$.

Transforming a uniform variable

Let U be uniform on $(0,1)$ and $D=10U$ cm. Find $P(D < 3 \text{ cm})$.

A nonlinear transformation

Let U be uniform on $(0,1)$ and $D=10\sqrt{U}$ cm. Find $P(D<3$ cm).

Sampling records from two sites

A hypothetical dataset has 80 records from site A and 20 from site B. Select one of the 100 records uniformly. Find the probability that it came from each site.

Selecting a site before selecting a record

A hypothetical dataset has 80 records from site A and 20 from site B. Select A or B with probability $1/2$ each, then uniformly within the selected site. Find the selection probability of one specified record from each site.

Accuracy and class proportions

A hypothetical test set contains 9,900 nondefective and 100 defective items. A system always predicts “nondefective.” Calculate its accuracy and defect detection fraction.

Practice: two fair dice

Assume two independent fair dice. List the 36 equally likely ordered outcomes and find the probability that the sum is 7.

Practice: at least one six

Assume two independent fair dice. Using the complement, find the probability of at least one six.

Review: describing a probability calculation

A hypothetical log has 50 requests, 7 above 200 ms. Select one request uniformly. Define the outcome, the latency random variable, and the event of latency above 200 ms; calculate its probability.

Review: assumptions behind the answer

For a probability calculation based on data or a transformed random variable, what must be specified about the experiment, event and distribution?

Further reading

Use the main notes for additional worked problems and the supplement for the formal development.



Speaker notes